Lim Jun Hao: Portfolio

Computer Science, Year 3. Interested in medical image analysis and explainable models.

Project 1. Diabetic retinopathy grading on the APTOS 2019 fundus dataset

Trained an EfficientNet-B3 grader in PyTorch on 3,662 retinal images, with class-weighted loss and test-time augmentation. Reached a quadratic weighted kappa of 0.89 on a held-out split, up from 0.81 for a ResNet-50 baseline. Added Grad-CAM saliency maps and checked them with a medical student, who confirmed the highlighted regions matched visible lesions in 41 of 50 cases. Measured calibration: expected calibration error fell from 9.2% to 3.1% after temperature scaling. Code and a 6-page report are on GitHub (limjunhao/aptos-grader).

Project 2. Kaggle: Retinal image quality classification

Finished in the top 8% (silver medal) of 1,150 teams, working alone. Wrote the data cleaning and augmentation pipeline and a short write-up of what failed.

Project 3. Final-year course project, NTU Neural Network and Deep Learning (SC4001)

Built a chest X-ray classifier with a group of four and led the evaluation. Our poster won the class's best-poster vote.

Experience

Summer attachment, data team of a local eye clinic (8 weeks): cleaned 12,000 anonymised screening records and built a dashboard the clinicians now use weekly. Reference available from the attachment supervisor on request.

Skills

PyTorch, scikit-learn, Grad-CAM and SHAP, experiment tracking, LaTeX and scientific writing.

Demo document for URECAMatch. Fictional student, not a real portfolio.